

FE-APP-01 - BSS Backoffice Web App

Version: v1.0

Status: Draft for review

Depends on:

- FE-APP-00-Frontend_Architecture_and_Shared_UX_Foundation-v1.0
- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- DD_API-00-API_Catalog_and_Gateway_Standards-v1.0
- DD_QA-01-End_to_End_Test_Strategy-v1.0
- FOUNDATION_AUTH
- FOUNDATION_FILE_STORAGE
- DD_TCK-01-Ticketing_and_Case_Management-v1.0
- All business DDs exposed through internal/backoffice APIs

1. Purpose

This DD defines the BSS Backoffice Web App.

The Backoffice is the main internal staff application for SOPHIX V3. It gives operational teams one place to search customers, manage customer-care work, monitor subscriptions, handle billing operations, supervise fulfillment, dispatch work orders, administer catalogs/configuration, review equipment/stock, and inspect operational incidents.

The Backoffice is Kenya-first but must remain operator and region configurable. The same codebase must support future deployments with different operators, countries, languages, currencies, themes, logos, tax labels, and feature flags.

2. Application Scope

The Backoffice is one deployable web application with role-based modules.

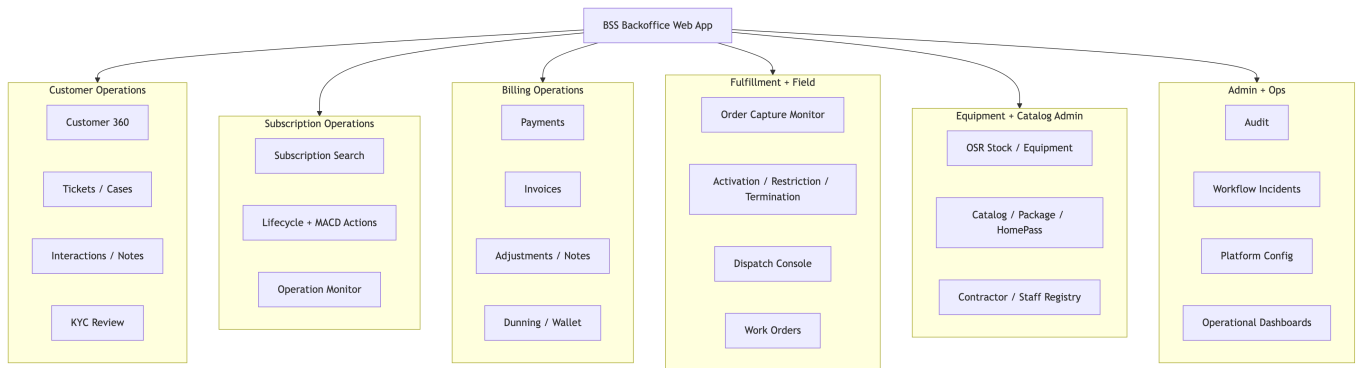
Included in this app:

Area	Included features	Primary module APIs used by the area
Customer operations	Customer 360, customer search, customer/account edit, KYC review, notes, interactions, preferences.	ILM Customer Service, FOUNDATION_FILE_STORAGE, NOT-01, ICN-01.
CRM / ticketing	Cases/tickets, interaction logging, task queues, escalations, attachments.	TCK-01, ILM Customer Service, W0-01, FOUNDATION_FILE_STORAGE, ICN-01.
Subscription operations	Subscription search, status, lifecycle actions, MACD actions, operation monitor.	SUB-LM-01, SUB-WF framework, SUB-WF-ACTIVATE-01, SUB-WF-PAUSE-01, SUB-WF-RESUME-01, SUB-WF-TERMINATE-01, SUB-WF-UPGRADE-01, SUB-WF-DOWNGRADE-01, SUB-WF-RESTRICT-01, SUB-WF-SUSPEND-NP-01, SUB-WF-RELOCATION-01, SUB-WF-MIGRATION-01.
Billing operations	Payment lookup, invoice lookup/download, adjustments, credit/debit notes, wallet, dunning queues.	BIL-01-PAY-01, BIL-02-READ-01, BIL-02-TAX-01, BIL-02-STA-01, BIL-02-ADJ-01, BIL-01-CN-01, BIL-03, BIL-04, BIL-05.
Fulfillment operations	Order capture monitor, activation/restriction/termination execution status, failure recovery view.	FUL-02, FUL-03, FUL-04, FUL-05, FUL-07, SUB-WF, BIL-01-PAY-01, W0-01.
Dispatch and work order	Work order queues, assignment, reassignment, escalation, finalization review.	W0-01, W0 flow satellite DDs, EM-02, ILM-CFG-02, OSR-INSTANCE-01, FOUNDATION_FILE_STORAGE.
OSR/equipment	Equipment instance lookup, stock movement view, swap/RMA queues, recovery status.	OSR-01, OSR-INSTANCE-01, OSR-RMA-01, PLM-CFG-06.
Catalog/config admin	Service catalog, package management, HomePass, tech region, tax/wallet/discount/equipment config.	PLM-CFG-01, SIP-01, RLM-CFG-01, ILM-CFG-02, PLM-CFG-02, PLM-CFG-03, PLM-CFG-04, PLM-CFG-05, PLM-CFG-06, PLM-CFG-08, BIL-CFG-01.

Area	Included features	Primary module APIs used by the area
Workforce admin	Contractor, staff, team, and agent registry views.	EM-02 Contractor and Staff Registry, EM-02 Contractor Team Agent Registry, ILM-CFG-02.
Platform operations	Audit search, workflow incidents, outbox/inbox health, operator config, admin tools.	Foundation admin APIs: auth/IAM, audit, outbox/inbox, cache, Camunda admin proxy, notification admin.
Operational dashboards	Embedded dashboards for daily queues. Strategic reports are covered by FE-APP-05.	Read APIs from the owning modules above; analytical reports use FE-APP-05 reporting APIs.

Illustration:

Figure 1 - Backoffice module map



3. Non-Scope

The Backoffice does not replace:

Application	Reason
Outdoor Sales / Franchise App	Sales agents need mobile-first lead/order/KYC/payment capture and offline drafts.
Contractor Technical Mobile App	Contractors need mobile job execution, scan/camera/evidence/offline support.
Customer Self-Care	Customers need simplified customer-facing language and self-service flows.
Reporting Portal	Management reporting can start embedded, but deep analytics/export is handled by FE-APP-05.

4. Users And Roles

Backoffice access is role-based. Navigation, screen actions, table actions, and high-risk buttons must be hidden or disabled according to role and feature flags. Backend authorization remains authoritative.

Role source:

- FOUNDATION_AUTH owns authentication, shared role names, JWT role claims, and Keycloak/LDAP group mapping.
- The Backoffice reads role claims from the OIDC token and uses them only to render navigation, menus, and enabled/disabled actions.
- Each module API remains responsible for endpoint authorization. A hidden button is not security.
- Modules must not invent local role names. If a Backoffice flow needs a new role, add it to FOUNDATION_AUTH and the Keycloak realm export first, then reference it in the owning module DD/API guard.

Role families are product-facing groupings used by the Backoffice UI. They are mapped to one or more FOUNDATION_AUTH roles and module-specific endpoint guards.

Role family	Typical permissions	Foundation/API link
Customer care agent	Search customer, view Customer 360, create ticket, add interaction/note, trigger allowed customer-care actions.	FOUNDATION_AUTH roles/claims; ILM/TCK/NOT/ICN endpoint role guards.
Customer care supervisor	Reassign tickets, approve escalations, override selected customer-care queues.	FOUNDATION_AUTH roles/claims; TCK/ILM supervisor-level endpoint guards.

Role family	Typical permissions	Foundation/API link
KYC reviewer L1/L2	Review KYC documents, approve/reject, request correction.	APPROVER_L1, APPROVER_L2 from FOUNDATION_AUTH; ILM/FUL KYC endpoint guards.
Subscription operator	View subscriptions and operations, trigger allowed lifecycle/MACD actions.	FOUNDATION_AUTH roles/claims; SUB-LM/SUB-WF endpoint role guards.
Billing operator	View invoices/payments/wallet, retry selected billing operations, create low-risk adjustments.	Billing roles from FOUNDATION_AUTH; BIL endpoint role guards.
Billing lead	Approve adjustments, write-offs, reversals, dunning overrides where configured.	BILLING_LEAD, DUNNING_ADMIN, WALLET_ADMIN, or configured billing roles from FOUNDATION_AUTH; BIL endpoint role guards.
Fulfillment operator	Monitor orders and fulfillment executions, retry failed activation/restriction/termination where allowed.	FOUNDATION_AUTH roles/claims; FUL endpoint role guards.
Dispatcher	Assign/reassign work orders, monitor field queues, handle route/slot conflicts.	FOUNDATION_AUTH roles/claims; WO/EM/ILM-CFG-02 endpoint role guards.
Work order supervisor	Cancel/escalate/reopen/review work orders, inspect finalization evidence.	SUPERVISOR or configured WO supervisor role from FOUNDATION_AUTH; WO endpoint role guards.
OSR/equipment operator	View stock/equipment, handle swaps/RMA/recovery queues.	FOUNDATION_AUTH roles/claims; OSR endpoint role guards.
Catalog admin	Manage service, package, HomePass, tax/wallet/discount/equipment config.	PRODUCT_MANAGER, BILLING_LEAD, or configured catalog roles from FOUNDATION_AUTH; PLM/SIP/RLM/BIL-CFG endpoint role guards.
Workforce admin	Manage contractors, staff, teams, agents, availability references.	FOUNDATION_AUTH roles/claims; EM endpoint role guards.
Platform operator	View workflow incidents, outbox/inbox, audit, health dashboards.	ADMIN, READ_ONLY, or configured platform roles from FOUNDATION_AUTH; foundation admin endpoint guards.
Super admin	Operator config, security-sensitive admin functions, system settings.	ADMIN from FOUNDATION_AUTH; platform/foundation admin endpoint guards.

Implementation rule: define a Backoffice role matrix in configuration as `roleFamily` -> `foundationRoles` -> `menu/actions`. This matrix controls UI visibility only. API calls must still include the bearer token and rely on backend role guards for final allow/deny decisions.

5. Navigation Model

Recommended top-level navigation:

Menu	Screens	Primary module API link
Home	My tasks, queue summaries, incidents, shortcuts, operator context.	ICN-01, module task APIs, SUB-WF, BIL-04, WO-01, FUL-03, OSR.
Customers	Customer search, Customer 360, KYC queues, interactions, preferences.	ILM Customer Service, FOUNDATION_FILE_STORAGE, NOT-01, ICN-01.
Tickets	Case/ticket queues, ticket detail, escalation, attachments.	TCK-01, WO-01, FOUNDATION_FILE_STORAGE, ICN-01.
Subscriptions	Subscription search, operation monitor, lifecycle/MACD actions.	SUB-LM-01, SUB-WF DDs.
Billing	Invoices, payments, wallet, adjustments, notes, dunning, tax invoice status.	BIL-01-PAY-01, BIL-02-READ-01, BIL-02-TAX-01, BIL-02-ADJ-01, BIL-01-CN-01, BIL-03, BIL-04, BIL-05.
Fulfillment	Order monitor, activation/restriction/termination status, retry queues.	FUL-02, FUL-03, FUL-04, FUL-05, FUL-07.
Work Orders	Dispatch board, WO queues, assignment, finalization review, support/shifting/install flows.	WO-01, WO flow satellite DDs, EM-02, ILM-CFG-02, OSR-INSTANCE-01.
Equipment	Stock balances, equipment instances, swaps, RMA, recovery.	OSR-01, OSR-INSTANCE-01, OSR-RMA-01, PLM-CFG-06.
Catalogs	Service catalog, packages, HomePass, tech regions, tax/wallet/discount/equipment config.	PLM-CFG-01, SIP-01, RLM-CFG-01, ILM-CFG-02, PLM-CFG-02, PLM-CFG-03, PLM-CFG-04, PLM-CFG-05, PLM-CFG-06, PLM-CFG-08, BIL-CFG-01.
Workforce	Contractors, teams, staff, availability, agent registry.	EM-02, ILM-CFG-02.
Reports	Operational dashboards and links to reporting portal.	Owning module read APIs for embedded views; reporting APIs in FE-APP-05.

Menu	Screens	Primary module API link
Admin	Audit, workflow incidents, outbox/inbox, users/roles links, operator config.	Foundation admin APIs: audit, IAM/auth, Camunda proxy, outbox/inbox, cache, config.

Navigation rules:

- Show only menus the user can access.
- Keep global search always available for staff roles that support customers.
- Preserve operator/country context visibly in the shell.
- Do not route users to empty screens because a feature flag is disabled.

6. Application Shell

Required shell components:

Component	Behavior	Module API/config link
Operator context	Shows current operator/country. Allows switch only for multi-operator users.	Runtime frontend config from FE-APP-00 and operator config APIs.
Global search	Search customer, account, subscription, invoice, payment, work order, equipment serial, ticket.	Federates to ILM, SUB-LM, BIL, W0, OSR, FUL, TCK search APIs.
Left navigation	Role-aware module navigation.	Keycloak/OIDC role claims plus feature flag/config APIs.
Task inbox	User tasks from KYC, tickets, billing approvals, dispatch, workflow incidents.	ICN-01, module task APIs, approved Camunda proxy.
Notification icon	Internal messages and system notices.	ICN-01, NOT-01.
Session profile	User, roles, language, logout.	FOUNDATION_AUTH OIDC profile/claims.
Environment marker	Visible for dev/UAT/staging. Hidden or minimal in production.	Runtime frontend config from FE-APP-00.
Breadcrumb	Required for deep detail screens.	Frontend route metadata; no backend call required.

7. Core Screen Groups

7.1 Home Dashboard

The home dashboard is role-aware.

Widgets:

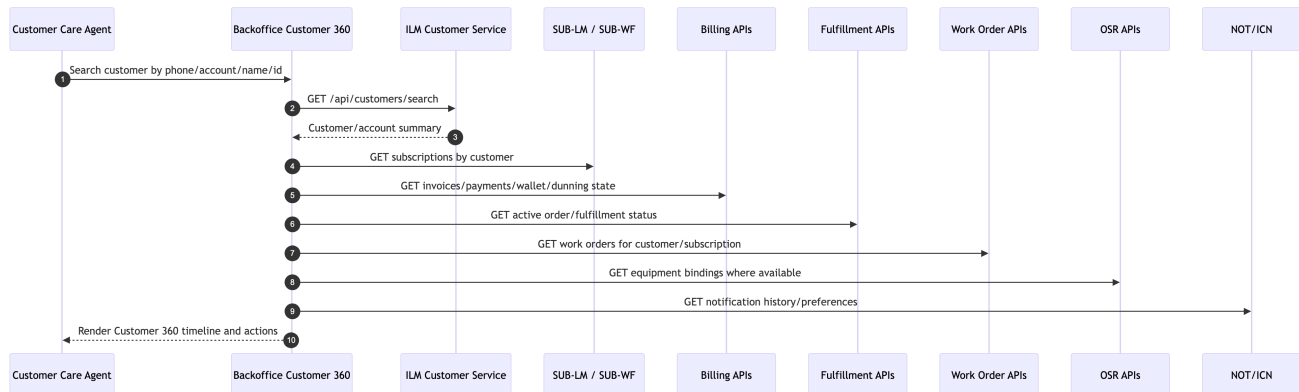
Widget	Shown to	Module API link	Representative API/event
My tasks	All operational users	ICN-01 and module user-task APIs exposed through Camunda wrappers.	GET /api/staff-notifications/inbox?status=DISPATCHED&limit=20; module-specific user-task APIs.
Stuck operations	Supervisors/platform operators	SUB-WF framework and approved Camunda incident/admin proxy.	GET /api/subscriptions/{subscription_id}/operations?since=...; incident list from admin proxy.
KYC pending	Customer care/KYC reviewers	ILM Customer Service, FUL-02.	KYC approval/rejection APIs and order user-task completion APIs.
Dunning risk	Billing teams	BIL-04.	GET /api/dunning-states?level={n}&status={s}&limit=50.
Work order backlog	Dispatch/work order teams	W0-01.	GET /api/work-orders?operatorCode=WIK&status=PENDING,ASSIGNED&limit=50.
Activation failures	Fulfillment teams	FUL-03, SUB-WF framework.	Activation status/retry APIs and subscription operation history APIs.
Stock exceptions	OSR teams	OSR-01, OSR-INSTANCE-01.	GET /api/stock-balances?...; equipment instance search APIs.

7.2 Customer Search And Customer 360

Customer 360 is the primary support screen.

Illustration:

Figure 2 - Customer 360 data assembly



Search keys:

- phone/MSISDN;
- account number;
- customer id;
- name;
- ID document number where authorized;
- subscription id;
- HomePass id;
- invoice/payment reference.

Customer 360 panels:

Panel	Module API link	Representative API
Customer profile/account	ILM Customer Service	GET /api/customers/{id}, GET /api/customer-accounts/{id}.
Contact methods/preferences	ILM Customer Service, NOT-01	GET /api/customers/{id}/contact-methods, GET /api/customers/{id}/notification-prefs.
KYC/documents	ILM Customer Service, FUL-02, FOUNDATION_FILE_STORAGE	KYC approval APIs and authorized document download/upload APIs.
Subscriptions	SUB-LM-01	GET /api/subscriptions?operatorCode=WIK&customerId={customer_id}.
Subscription operations	SUB-WF framework	GET /api/subscriptions/{subscription_id}/operations?since=...
Billing summary	BIL-02-READ-01, BIL-01-PAY-01, BIL-05, BIL-04	Invoice, payment, wallet, and dunning state APIs.
Active orders/fulfillment	FUL-02, FUL-03, FUL-04, FUL-05	Order detail/monitor APIs and execution status APIs.
Work orders	W0-01	GET /api/work-orders?customerId=...
Equipment	OSR-INSTANCE-01, OSR-01	Equipment instance and stock/location APIs.
Notifications	NOT-01, ICN-01	Notification history, preference, and staff inbox APIs.
Tickets/cases	TCK-01	GET /api/tickets?operatorCode=WIK&customerId={customer_id}.
Interactions/notes	ILM Customer Service, TCK-01	Customer notes/interactions APIs and ticket timeline/comment APIs.
Audit timeline	Local module audit, central audit/index if available	GET /api/audit?... or module-specific audit APIs.

Rules:

- Customer 360 may aggregate data, but must not create hidden write ownership.

- Each panel must show stale/error states independently.
- If one backend fails, the rest of Customer 360 should still render where possible.
- High-risk actions must show source module and expected impact.
- Initial load must be small: customer identity/header, active subscriptions, balance/debt summary, open tickets, and recent timeline only.
- History-heavy panels such as invoices, work orders, equipment, notifications, and audit load on tab open or lazy load.
- If direct gateway calls create unacceptable latency, introduce a Backoffice BFF/read aggregator for Customer 360 reads only. The BFF must not own business state or execute writes across modules.

7.3 Ticketing / CRM Cases

Ticketing supports customer-care operations through DD_TCK-01 Ticketing and Case Management.

TCK owns case lifecycle, SLA, assignment, ticket comments, attachment metadata, linked entities, and ticket-to-WO creation. ILM still owns customer profile, contact methods, and general customer notes/interactions. WO owns field execution after TCK creates a Work Order.

Ticket capabilities:

Feature	Description	Module API link	Representative API
Create case	From Customer 360 or global menu.	TCK-01.	POST /api/tickets.
Link case	Link customer, account, subscription, invoice, work order, equipment serial.	TCK-01; linked entity detail stays in owning module.	GET /api/tickets/{ticket_id}; linked entity panel calls ILM/SUB/BIL/WO/OSR APIs as needed.
Categorize	Complaint, billing, technical, sales, KYC, relocation, termination, general inquiry.	TCK-01.	Ticket category catalog APIs from TCK.
Assign	User/team assignment and queue movement.	TCK-01, optionally EM-02 for team/staff display.	POST /api/tickets/{ticket_id}/assign.
SLA	Priority, due date, escalation state.	TCK-01.	Ticket detail and category/SLA config APIs.
Timeline	Interactions, notes, status changes, attachments.	TCK-01, FOUNDATION_FILE_STORAGE.	GET /api/tickets/{ticket_id}, POST /api/tickets/{ticket_id}/comments.
Resolve	Resolution code, customer communication, follow-up flag.	TCK-01, NOT-01 if customer notification is triggered by backend.	POST /api/tickets/{ticket_id}/resolve.
Create Work Order	For technical/support categories where TCK category config allows field intervention.	TCK-01, WO-01.	POST /api/tickets/{ticket_id}/work-orders; TCK calls POST /api/work-orders.

8. Subscription Operations

Subscription screens must make the SUB-LM/SUB-WF split visible.

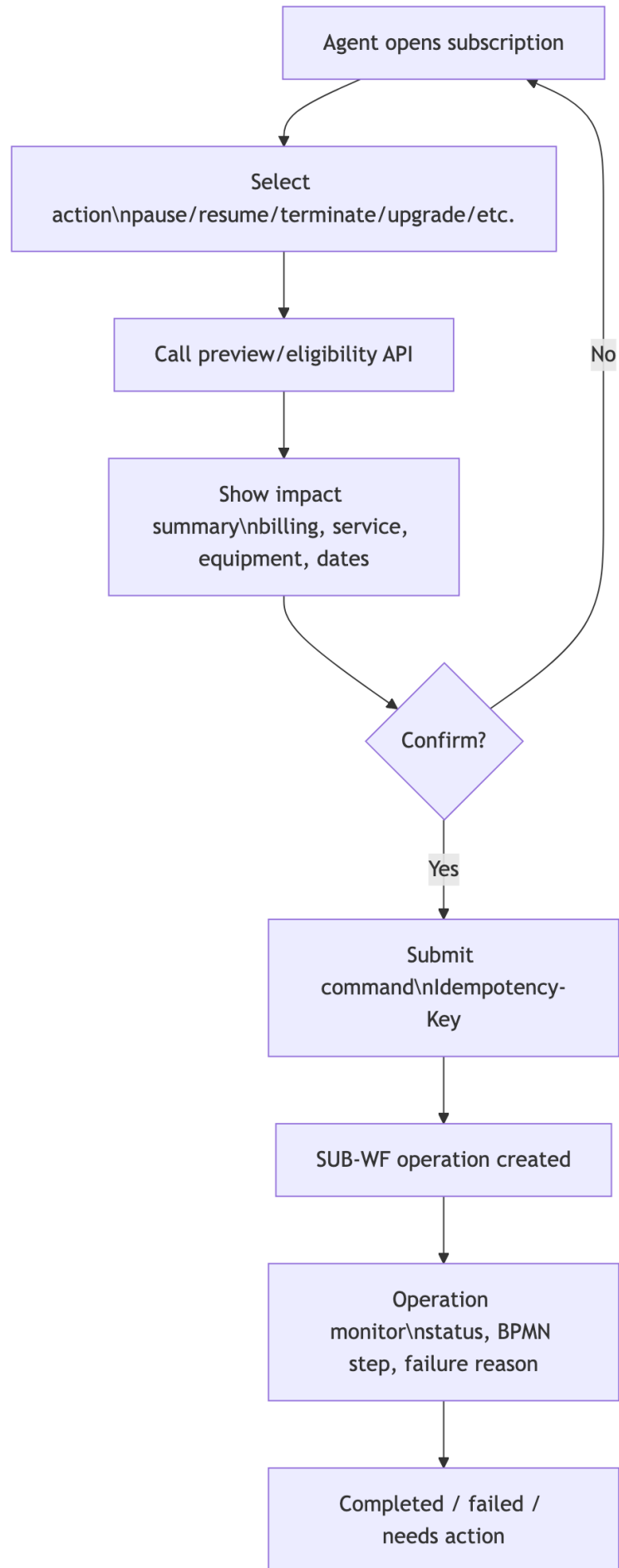
Screen groups:

Screen	Purpose	Module API link	Representative API
Subscription search/list	Find subscriptions by customer, HomePass, package, status, operator.	SUB-LM-01.	GET /api/subscriptions?operatorCode=WIK&customerId=...&statusCode=ACTIVE&page=0&size=50.
Subscription detail	Show SUB-LM master data, status, cycle fields, restrictions, linked customer/account.	SUB-LM-01, ILM Customer Service.	GET /api/subscriptions/{subscription_id} plus ILM customer/account detail APIs.
Operation history	Show SUB-WF operations, status, actor, BPMN process id, failure reason.	SUB-WF framework.	GET /api/subscriptions/{subscription_id}/operations?since=...
Lifecycle actions	Activate, pause, resume, terminate where role/config allows.	SUB-WF-ACTIVATE-01, SUB-WF-PAUSE-01, SUB-WF-RESUME-01, SUB-WF-TERMINATE-01.	POST /api/subscriptions/{subscription_id}/activate, pause, resume, terminate.

Screen	Purpose	Module API link	Representative API
MACD actions	Upgrade, downgrade, restrict, suspend, relocation, migration where role/config allows.	SUB-WF-UPGRADE-01, SUB-WF-DOWNGRADE-01, SUB-WF-RESTRICT-01, SUB-WF-SUSPEND-NP-01, SUB-WF-RELOCATION-01, SUB-WF-MIGRATION-01.	Command APIs under /api/subscriptions/{subscription_id}/.. ..
Operation monitor	Track in-flight and failed operations across queues.	SUB-WF framework, approved Camunda proxy.	GET /api/subscriptions/{subscription_id}/in-flight-operation; operation list APIs.

Illustration:

Figure 3 - Subscription action flow



Action rules:

- Always call preview/eligibility endpoint before high-impact command if available.
- Show impact summary before submit: billing, service, equipment, customer notification, dates.
- Commands must include idempotency key.
- After submit, show operation status, not just "success".
- For long-running operations, provide a link to operation monitor.
- Failed operations must show safe retry/manual recovery actions only when backend exposes them.

9. Billing Operations

Billing screens are high-risk and must be role-gated.

Screen groups:

Screen	Purpose	Module API link	Representative API
Payment search/detail	Search payment, inspect allocation/reversal, view receipt context.	BIL-01-PAY-01.	POST /api/payments/receive, POST /api/payments/{payment_id}/reverse, payment lookup APIs.
Invoice search/detail	Search invoice, view invoice status, lines, taxes, balances.	BIL-02, BIL-02-READ-01.	GET /api/invoices, GET /api/invoices/{id}, GET /api/invoices/{id}/status.
Invoice PDF download	Download authorized customer invoice PDF.	BIL-02-READ-01, FOUNDATION_FILE_STORAGE.	GET /api/invoices/{id}/pdf.
Tax invoice status/retry	Monitor tax signing/issuance/cancellation.	BIL-02-TAX-01.	POST /api/tax-invoices/manual, POST /api/tax-invoices/{id}/retry-signing, POST /api/tax-invoices/{id}/cancel.
Invoice status recovery	Controlled invoice status repair.	BIL-02-STA-01.	Invoice status recovery APIs from the DD.
Adjustments	Dispute, adjustment, write-off, correction flows.	BIL-02-ADJ-01.	POST /api/disputes, POST /api/admin/disputes/{id}/resolve, POST /api/admin/invoices/{id}/write-off.
Notes	Credit/debit note list, detail, and application.	BIL-01-CN-01.	GET /api/invoices/credit-notes, GET /api/invoices/debit-notes, note application APIs.
Cycle close monitor	Cycle close run, failed evaluation, forced close where authorized.	BIL-03.	GET /api/cycle-close-runs/{run_id}, GET /api/cycle-evaluations/failed, POST /api/cycle-evaluations/{subscriptionId}/force-close.
Dunning queues	Debt escalation queues, termination review, dunning history.	BIL-04.	GET /api/dunning-states?level={n}&status={s}&limit=50, GET /api/dunning-states/{subscriptionId}/history.
Wallet/top-up	Wallet balance, transactions, top-up channel visibility.	BIL-05.	GET /api/wallets/{subscriptionId}, GET /api/wallets/{subscriptionId}/balance, GET /api/wallets/{subscriptionId}/transactions?limit=50.

Required UX:

- Money values must show currency from operator config.
- Show gross/net/tax breakdown where API provides it.
- Show legal invoice number separately from internal id.
- Show payment allocation and reversals clearly.
- High-risk actions require confirmation and reason.
- Approval-required actions must show pending approval state.

- Downloaded PDFs must go through authorized app/API, not direct public object URLs.

10. Fulfillment Operations

Fulfillment screens focus on order capture and service execution.

Screen groups:

Screen	Purpose	Module API link	Representative API/event
Order monitor	Track FUL-02 orders by stage: capture, validate, KYC, payment, install, subscription, activation, cancellation.	FUL - 02.	Order query/detail APIs, POST /api/orders/{id}/cancel.
Order detail	Show order data, customer, package, HomePass, payment, WO, subscription, activation status.	FUL - 02, ILM, SIP - 01, RLM - CFG - 01, BIL - 01 - PAY - 01, W0 - 01, SUB - LM - 01, FUL - 03.	Load each panel from its owner API; do not denormalize ownership in the frontend.
Activation monitor	Track FUL-03 service activation attempts and failures.	FUL - 03, SUB - WF framework.	POST /api/activation/execute and activation retry/status APIs.
Restriction monitor	Track FUL-04 restriction/unrestriction execution.	FUL - 04.	Restriction execution status APIs/events.
Termination monitor	Track FUL-05 termination execution.	FUL - 05.	Termination execution status APIs/events.
HomePass lifecycle	Track FUL-07/RLM lifecycle flows where exposed.	FUL - 07, RLM - CFG - 01.	POST /api/homepasses/{id}/transitions/{process_id}/approve, HomePass detail APIs.

Rules:

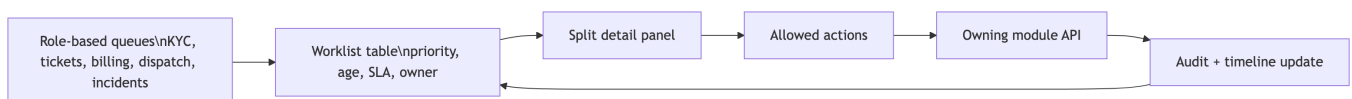
- Show Camunda/message-wait states clearly: waiting for payment, waiting for WO finalization, waiting for callback.
- Retry buttons must call backend recovery APIs only.
- Do not let Backoffice manually skip required KYC/payment/install checks unless a DD explicitly defines an override.

11. Dispatch And Work Orders

Work order screens support dispatchers and supervisors.

Illustration:

Figure 4 - Backoffice work queue pattern



Screen groups:

Screen	Purpose	Module API link	Representative API
Dispatch board	Work order map/list/calendar grouped by region, skill, contractor, status.	W0 - 01, ILM - CFG - 02, EM - 02.	GET /api/work-orders?... , tech region APIs, contractor/staff lookup APIs.
WO queue	Filterable list of pending, assigned, in-progress, finalization-pending, completed, cancelled.	W0 - 01.	GET /api/work-orders?operatorCode=WIK&status=...
WO detail	Customer, location, subscription, task type, phase, assignment, notes, attachments, equipment.	W0 - 01, ILM, SUB - LM - 01, OSR - INSTANCE - 01, FOUNDATION_FILE_STORAGE.	GET /api/work-orders/{work_order_id} plus owner panel APIs.
Assignment	Assign/reassign to contractor/team/technician.	W0 - 01, EM - 02, ILM - CFG - 02.	POST /api/work-orders/{id}/assign, POST /api/work-orders/{id}/claim.
Finalization review	Inspect checklist, photos, serials, signatures, failure reasons.	W0 - 01 flow satellite DDS, OSR - INSTANCE - 01, FOUNDATION_FILE_STORAGE.	POST /api/work-orders/{id}/finalize-first-confirm and flow-specific finalize APIs.
Escalation	Create support/QCS/escalation path where allowed.	W0 - 01, TCK - 01 when escalation becomes a ticket.	WO escalation/support APIs or POST /api/tickets.

Rules:

- Assignment must validate contractor/staff/team through EM APIs.
- Equipment serial/binding data must come from OSR.
- Photos/evidence must be streamed through authorized file APIs.
- Reassignment/cancellation must show downstream impact.
- Multi-phase work orders must show current phase and next phase.

12. OSR And Equipment

Equipment screens are operationally sensitive.

Screen groups:

Screen	Purpose	Module API link	Representative API
Stock balances	View stock by operator, location, SKU, contractor/team.	OSR-01, PLM-CFG-06.	GET /api/stock-balances?operatorCode=WIK&locationId=...; equipment SKU APIs.
Stock movements	Movement history and reason codes.	OSR-01.	GET /api/stock-movements?... GET /api/stock-reason-codes?...
Equipment instance search	Search by serial, instance id, state, customer/subscription, contractor.	OSR-INSTANCE-01.	GET /api/equipment-instances?operator=WIK&state=...
Equipment lifecycle	Show lifecycle events and current holder/location.	OSR-INSTANCE-01.	GET /api/equipment-instances/{instance_id}/lifecycle-events.
Swap/RMA queues	Track GPON/HFC/EQP/EQU flows.	OSR-RMA-01.	GET /api/swap-requests?... POST /api/swap-requests/gpon.
Recovery exceptions	Equipment not recovered, mismatched serial, damaged, RMA pending.	OSR-INSTANCE-01, OSR-RMA-01, WO-01.	Equipment state/search APIs, swap/RMA queue APIs, linked WO APIs.

Rules:

- No equipment state changes should be performed by UI-only logic.
- All state changes call OSR APIs.
- Serial scan/manual entry must validate against OSR.
- Show audit trail for every equipment state transition.

13. Catalog And Configuration Admin

Catalog/config screens support operator setup and controlled changes.

Screen groups:

Area	Module API link	Representative API
Service catalog	PLM-CFG-01.	GET /api/services/by-code/{service_code} and service list/create/update APIs.
Package management	SIP-01.	GET /api/packages/{package_id} and package list/create/update APIs.
HomePass configuration	RLM-CFG-01.	GET /api/homepasses/{homepass_id}, POST /api/homepasses/bulk-import.
Tech region registry	ILM-CFG-02.	GET /api/tech-regions?operatorCode=WIK&active=true, PATCH /api/tech-regions/{id}.
Tax configuration	PLM-CFG-02.	POST /api/tax/admin/rules, POST /api/tax/compute.
Wallet configuration	PLM-CFG-03.	Wallet config and wallet channel config APIs.
Discount catalog	PLM-CFG-04.	Discount catalog list/create/update APIs.

Area	Module API link	Representative API
Adjustment catalog/type	PLM-CFG-05, PLM-CFG-08.	GET /api/adjustment-reason-codes?operatorCode=WIK&active=true, create/update APIs.
Equipment type catalog	PLM-CFG-06.	GET /api/equipment-kinds, GET /api/equipment-skus.
Billable event catalog	BIL-CFG-01.	POST /api/billable-events and billable event catalog APIs.

Required UX:

- List/detail/edit with version/status where the DD supports lifecycle.
- Show active/inactive/retired state clearly.
- Validate effective dates.
- Warn when a change affects active packages/subscriptions/billing.
- Prefer preview/validate APIs before publishing high-impact config.
- Keep audit trail visible.

14. Workforce Admin

Workforce admin screens use EM DDs.

Features:

Feature	Module API link	Representative API
Contractor registry	EM-02 Contractor and Staff Registry.	GET /api/contractors/{contractor_id} and contractor list/create/update APIs.
Staff registry	EM-02 Contractor and Staff Registry.	GET /api/contractor-staff/{staff_id}, GET /api/contractor-staff?contractorId=X&active=true.
Team registry	EM-02 Contractor Team Agent Registry.	Team registry APIs from EM team/agent DD.
Agent mapping	EM-02 Contractor Team Agent Registry.	Team-agent mapping APIs from EM team/agent DD.
Availability/slot commitments	EM-02.	POST /api/em/contractor-availability, GET /api/contractor-slot-commitments?slotId=X&date=Y.
Skill/region assignment references	EM-02, ILM-CFG-02.	Contractor/staff lookup APIs and tech region candidate group APIs.
Active/inactive lifecycle	EM-02.	Contractor/staff lifecycle update APIs from EM DD.

Rules:

- WO assignment uses EM data; do not duplicate workforce records in WO frontend state.
- Inactive staff/contractors must remain visible in history but not selectable for new assignment.

15. Platform Admin And Operations

Platform admin screens are for operational support, not normal business users.

Features:

Screen	Purpose	Module API link	Representative API
Audit search	Search by actor, entity, customer, time range.	Central audit if enabled; otherwise local module audit APIs.	GET /api/audit?... or module audit endpoints.
Camunda incidents	View incidents, failed tasks, timers, process instance ids.	Approved Camunda admin proxy plus owning module wrappers.	Incident/task APIs exposed through backend admin proxy.
Outbox monitor	Stuck unpublished outbox rows by module.	Foundation outbox admin APIs.	POST /admin/outbox/replay; outbox list/status APIs.
Inbox monitor	Failed consumer rows and retry controls.	Foundation inbox admin APIs.	POST /admin/inbox/retry/{eventId}; inbox list/status APIs.
Cache admin	Invalidate/reconcile cache where foundation APIs allow it.	Foundation cache admin APIs.	POST /admin/cache/invalidate, cache stats/reconcile APIs.

Screen	Purpose	Module API link	Representative API
Operator config	View runtime frontend/operator config.	Frontend config service and operator config APIs.	Runtime config endpoint from FE-APP-00.
Feature flags	View flags; editing depends on platform policy.	Feature flag/config service.	Feature flag read APIs; edit only if policy exposes admin APIs.
Health dashboard	API, DB, Kafka, Redis, Camunda, Drools status.	Foundation/platform health APIs and module health endpoints.	/health, /ready, and platform dependency health APIs.

Rules:

- Dangerous repair actions require elevated role and confirmation.
- All repair actions must require reason/comment.
- Show correlation ids and operation ids.
- Prefer backend runbook APIs. Do not implement repair logic in the frontend.

16. Global Search

Global search should support:

Search type	Module API link	Representative API
Customer phone/name/id	ILM Customer Service.	GET /api/customers/search?key=msisdn&value=...
Account id	ILM Customer Service.	GET /api/customer-accounts/{id}.
Subscription id	SUB-LM-01.	GET /api/subscriptions/{subscription_id}.
Invoice number/id	BIL-02-READ-01.	GET /api/invoices/{id} or invoice search API.
Payment reference	BIL-01-PAY-01.	Payment lookup APIs from DD.
Work order id	W0-01.	GET /api/work-orders/{work_order_id}.
Equipment serial	OSR-INSTANCE-01.	GET /api/equipment-instances?operator=WIK&serial=...
Order id	FUL-02.	Order detail/query APIs.
Ticket/case id	TCK-01.	GET /api/tickets/{ticket_id}.

Search results must show type, status, customer reference where allowed, and direct action/view links.

17. Common Work Queue Pattern

A work queue is a screen pattern for lists of items that require human review, assignment, retry, approval, or follow-up.

It is not one shared backend table. Each queue is owned by the module that owns the business object. The Backoffice provides a consistent UI pattern across modules, but it must call the owning module API for the queue data and actions.

Examples:

Queue	Owning module/API	What the user does
KYC pending L1/L2	ILM Customer Service, FUL-02 task APIs	Review documents, approve, reject, request correction.
Tickets waiting assignment	TCK-01	Assign ticket to user/team, escalate, add comment.
Failed subscription operations	SUB-WF framework, approved Camunda proxy	Inspect operation failure, retry only through exposed recovery action.
Activation failures	FUL-03, SUB-WF framework	Review activation error, trigger backend retry if allowed.
Dunning review	BIL-04	Review debt state, approve configured dunning override or termination review.
Tax invoice failures	BIL-02-TAX-01	Retry signing/cancellation where allowed.
Work orders pending assignment	W0-01, EM-02, ILM-CFG-02	Assign/reassign contractor/team/technician.
Equipment recovery exceptions	OSR-INSTANCE-01, OSR-RMA-01, W0-01	Investigate missing/damaged/mismatched equipment and open follow-up actions.

Queue	Owning module/API	What the user does
Outbox/inbox failures	Foundation outbox/inbox admin APIs	Inspect correlation id, retry through backend admin endpoint.

Expected API shape:

```
GET /api/{module-owned-resource}?status=...&assignee=...&team=...&priority=...&ageBucket=...&page=0&size=50
Authorization: Bearer <jwt>
```

Expected list response shape:

```
{
  "items": [
    {
      "id": "TCK-2026-000123",
      "type": "TECHNICAL_SUPPORT",
      "status": "OPEN",
      "priority": "HIGH",
      "ownerUserId": "agent-101",
      "ownerTeamId": "care-l2",
      "slaDueAt": "2026-05-29T12:00:00Z",
      "customerId": "CUS-100045",
      "summary": "Customer reports no internet after installation"
    }
  ],
  "page": 0,
  "size": 50,
  "totalElements": 231
}
```

Queue UI requirements:

- show saved filters where the owning API supports them;
- support role/team queues from FOUNDATION_AUTH role claims and module assignment fields;
- show SLA/age indicators, priority, owner/assignee, and status;
- use split list/detail mode for fast triage;
- show refresh time and stale-data state;
- allow bulk select only where the owning module API exposes a safe bulk action;
- allow export only for authorized users and only through an owning module/reporting API;
- route every row action to the owning module API, not to local frontend logic.

The frontend may implement one reusable queue component, but each queue configuration must declare:

```
{
  "queueCode": "tickets_waiting_assignment",
  "ownerModule": "TCK-01",
  "listApi": "GET /api/tickets?status=OPEN,TRIAGED&assignee=UNASSIGNED",
  "detailRoute": "/tickets/:ticketId",
  "allowedActions": ["assign", "comment", "escalate"],
  "requiredRolesFromFoundationAuth": ["SUPERVISOR"]
}
```

18. Frontend To Module API Map

This section links Backoffice feature groups to the actual module APIs defined in the DDs.

Rules:

- The Backoffice calls module APIs through the approved API gateway/base URL.
- The Backoffice must not call another module's database.
- Endpoints below are representative contracts from the DDs. Exact query parameters should follow the owning DD/API implementation.
- If a screen needs data from several modules, load each panel from its owning API and show panel-level loading/error state.

18.1 Home Dashboard APIs

Widget	Owning module/DD	Representative APIs/events	Frontend use
My tasks	ICN, Camunda-backed module task APIs	GET /api/staff-notifications/inbox?status=DISPATCHED&limit=20, module-specific user-task APIs	Show user's actionable tasks and acknowledgements.
Stuck subscription operations	SUB-WF framework, Camunda	GET /api/subscriptions/{subscription_id}/in-flight-operation, GET /api/subscriptions/{subscription_id}/operations?since=...	Show stuck/failed operations and link to operation detail.
KYC pending	ILM Customer Service, FUL-02 KYC	POST /api/customers/{id}/kyc/l1-approve, POST /api/customers/{id}/kyc/final-approve, order user-task completion APIs	Show KYC queues and approval tasks.
Dunning risk	BIL-04	GET /api/dunning-states?level={n}&status={s}&limit=50, GET /api/dunning-states/pending-termination-review	Show debt escalation queues.
Work order backlog	WO-01	GET /api/work-orders?operatorCode=WIK&status=PENDING, ASSIGNED&kind=SUPPORT&techRegionCode=...&limit=50	Show field work backlog.
Activation failures	FUL-03, SUB-WF	FUL activation status APIs, SUB-WF operation history APIs	Show failed activation/retry queue.
Stock exceptions	OSR	GET /api/stock-balances?operatorCode=WIK&locationId=..., equipment instance queries	Show stock/equipment exceptions.

18.2 Customer Search And Customer 360 APIs

Panel/action	Owning module/DD	Representative APIs	Frontend use
Customer search	ILM-CFG-01 Customer Service	GET /api/customers/search?key=msisdn&value=%2B254712345678&limit=10, GET /api/customers/search	Global/customer search.
Customer profile	ILM-CFG-01	GET /api/customers/{id}, PATCH /api/customers/{id}	Identity, profile, allowed profile edits.
Customer account	ILM-CFG-01	GET /api/customer-accounts/{id}, GET /api/customer-accounts?customerId=..., PATCH /api/customer-accounts/{id}	Account status, account fields.
Contact methods	ILM-CFG-01	GET /api/customers/{id}/contact-methods, POST /api/customers/{id}/contact-methods, PATCH /api/contact-methods/{id}, POST /api/contact-methods/{id}/verify	Phone/email/address contact management.
KYC approvals	ILM-CFG-01, FUL-02 KYC	POST /api/customers/{id}/kyc/l1-approve, POST /api/customers/{id}/kyc/final-approve, POST /api/customers/{id}/kyc/reject	KYC review actions.
Notes/interactions	ILM-CFG-01	GET /api/customers/{id}/notes?limit=20, POST /api/customers/{id}/notes, GET /api/customers/{id}/interactions?limit=20, POST /api/customers/{id}/interactions	Customer care timeline.
Notification preferences	ILM-CFG-01, NOT-01	GET /api/customers/{id}/notification-prefs, PUT /api/customer-accounts/{id}/notification-pref	View/update communication preferences.
Subscriptions panel	SUB-LM-01	GET /api/subscriptions?operatorCode=WIK&customerId={customer_id}&statusCode=ACTIVE&page=0&size=50, GET /api/subscriptions/{subscription_id}	List customer subscriptions and open detail.

Panel/action	Owning module/DD	Representative APIs	Frontend use
Operation history panel	SUB-WF framework	GET /api/subscriptions/{subscription_id}/operations?sinc e=..., GET /api/subscriptions /{subscription_id}/in-flight -operation	Show active and past lifecycle/MACD operations.
Invoices panel	BIL-02-READ-01	GET /api/invoices?account_id={account_id}, GET /api/invoices/{id}, GET /api/invoices/{id}/pdf	Invoice list/detail/PDF.
Wallet panel	BIL-05	GET /api/wallets/{subscriptionId}, GET /api/wallets/{subscriptionId}/balance, GET /api /wallets/{subscriptionId}/tr ansactions?limit=50	Wallet status and transaction history.
Dunning panel	BIL-04	GET /api/dunning-states/{sub scriptionId}, GET /api/dunnin g-states/{subscriptionId}/hi story	Debt/dunning state.
Work orders panel	WO-01	GET /api/work-orders?custome rId=..., GET /api/work-orders /{work_order_id}	Field visit history.
Equipment panel	OSR-INSTANCE-01	GET /api/equipment-instances /{instance_id}, GET /api/equi pment-instances?operator=WIK &state=...	Bound/recovered equipment visibility.
Notifications panel	NOT-01, ICN-01	Notification history/preference APIs from NOT/ICN	Show customer/staff communications.
Tickets panel	TCK-01	GET /api/tickets?operatorCod e=WIK&customerId={customer_i d}&page=0&size=50	Customer support/case history.

18.3 Ticketing / CRM APIs

Feature	Owning module/DD	Representative APIs	Frontend use
Ticket search/list	TCK-01	GET /api/tickets?operatorCod e=WIK&customerId=...&status= OPEN, ASSIGNED&page=0&size=50, GET /api/tickets?linkedEntit yType=WORK_ORDER&linkedEntit yId=...	Ticket queues, Customer 360 ticket panel, global ticket search.
Ticket detail	TCK-01	GET /api/tickets/{ticket_id}	Case detail, timeline, linked entities, SLA, status.
Create ticket	TCK-01	POST /api/tickets	Create support/complaint/follow-up case.
Assign ticket	TCK-01	POST /api/tickets/{ticket_id }/assign	Move to queue/user/team.
Ticket comments	TCK-01	POST /api/tickets/{ticket_id }/comments	Add internal or customer-visible comment.
Ticket attachments	TCK-01 + FILE_STORAGE	TCK attachment APIs backed by file storage	Upload/list ticket attachments.
Create WO from ticket	TCK-01 + WO-01	POST /api/tickets/{ticket_id }/work-orders; TCK calls WO POST /api/work-orders	Dispatch field work from support ticket.
Resolve/reopen ticket	TCK-01	POST /api/tickets/{ticket_id }/resolve, POST /api/tickets/ {ticket_id}/reopen	Close or reopen case.
Customer notes	ILM-CFG-01	GET /api/customers/{id}/note s?limit=20, POST /api/customers/{id}/notes, PATCH /api/notes/{id}/supersede	General customer notes outside a formal ticket.
Interactions	ILM-CFG-01	GET /api/customers/{id}/inte ractions?limit=20, POST /api/ customers/{id}/interactions	Log calls, visits, chat, complaint interactions.

Feature	Owning module/DD	Representative APIs	Frontend use
Staff notification/task	ICN-01, NOT-01	GET /api/staff-notifications/inbox?status=DISPATCHED&limit=20, POST /api/staff-notifications/{notification_id}/ack	Agent task and internal communication handling.

Backoffice must not create a frontend-only ticket database. If a customer issue needs case lifecycle, use TCK. If the case needs field work, use TCK's create-WO-from-ticket API so the ticket timeline and WO link remain auditable.

18.4 Subscription Operations APIs

Screen/action	Owning module/DD	Representative APIs	Frontend use
Subscription list/detail	SUB-LM-01	GET /api/subscriptions?operatorCode=WIK&customerId=...&statusCode=ACTIVE&page=0&size=50, GET /api/subscriptions/{subscription_id}	Search and display master subscription state.
Create subscription	SUB-LM-01	POST /api/subscriptions	Backoffice/manual create only where role and flow allow.
Activation preview/start	SUB-WF-ACTIVATE-01	POST /api/subscriptions/{subscription_id}/activate/preview, POST /api/subscriptions/{subscription_id}/activate	Activation action from detail screen.
Pause	SUB-WF-PAUSE-01	POST /api/subscriptions/{subscription_id}/pause, GET /api/subscriptions/{subscription_id}/pause-history	Customer/operator pause.
Resume	SUB-WF-RESUME-01	POST /api/subscriptions/{subscription_id}/resume	Resume eligible subscription.
Terminate	SUB-WF-TERMINATE-01	POST /api/subscriptions/{subscription_id}/terminate/preview, POST /api/subscriptions/{subscription_id}/terminate	Termination impact/confirm flow.
Upgrade	SUB-WF-UPGRADE-01	POST /api/subscriptions/{subscription_id}/upgrade, DELETE /api/subscriptions/{subscription_id}/pending-upgrade/{operation_id}	Upgrade and cancel future upgrade.
Downgrade	SUB-WF-DOWNGRADE-01	POST /api/subscriptions/{subscription_id}/downgrade, DELETE /api/subscriptions/{subscription_id}/pending-downgrade/{operation_id}	Downgrade and cancel future downgrade.
Restrict/unrestrict	SUB-WF-RESTRICT-01	GET /api/subscriptions/{subscription_id}/restrictions, POST /api/subscriptions/{subscription_id}/restrictions, DELETE /api/subscriptions/{subscription_id}/restrictions/{code}	Add/remove partial restrictions.
Suspend non-payment	SUB-WF-SUSPEND-NP-01	POST /api/subscriptions/{subscription_id}/suspend-np	Usually BIL-04-driven; Backoffice only if role/config allows.
Relocation	SUB-WF-RELOCATION-01	POST /api/subscriptions/{subscription_id}/relocate	Move subscription to target HomePass.
Migration	SUB-WF-MIGRATION-01	POST /api/subscriptions/{subscription_id}/migrate	Compound package/HomePass/technology migration.
Operation monitor	SUB-WF framework	GET /api/subscriptions/{subscription_id}/operations?since=..., GET /api/subscriptions/{subscription_id}/in-flight-operation	Track operation status, failure, retry links.

For every command endpoint, Backoffice must send Idempotency-Key and show the returned operation id/status.

18.5 Billing Operations APIs

Screen/action	Owning module/DD	Representative APIs	Frontend use
Invoice search/list	BIL-02-READ-01	GET /api/invoices, GET /api/invoices?account_id=X&status=PENDING&sort=issue_date_desc&limit=20, GET /api/invoices?operator_code=WIK&customer_id=...	Invoice queues and account/customer invoice list.
Invoice detail	BIL-02-READ-01	GET /api/invoices/{id}, GET /api/invoices/{id}/status	Invoice detail and status panel.
Invoice PDF	BIL-02-READ-01 + FILE_STORAGE	GET /api/invoices/{id}/pdf	Authorized PDF download.
Tax invoices	BIL-02-TAX-01	POST /api/tax-invoices/manual, POST /api/tax-invoices/{id}/retry-signing, POST /api/tax-invoices/{id}/cancel, POST /api/tax-invoices/{id}/cancel/approve	Tax invoice operations and recovery.
Payments	BIL-01-PAY-01	POST /api/payments/receive, POST /api/payments/{payment_id}/reverse, payment lookup APIs	Payment application, reversal, receipt view.
Adjustments	BIL-02-ADJ-01	POST /api/disputes, POST /api/admin/disputes/{id}/resolve, POST /api/admin/invoices/{id}/write-off, adjustment APIs	Adjustment, dispute, write-off flows.
Credit/debit notes	BIL-01-CN-01	GET /api/invoices/credit-notes, GET /api/invoices/debit-notes, note application APIs	Notes list/detail and application.
Cycle close	BIL-03	GET /api/cycle-close-runs/{run_id}, GET /api/cycle-close-runs?limit=50, GET /api/cycle-evaluations/failed, POST /api/cycle-evaluations/{subscriptionId}/force-close	Cycle close monitoring and controlled repair.
Dunning	BIL-04	GET /api/dunning-programs, GET /api/dunning-states?level={n}&status={s}&limit=50, GET /api/dunning-states/{subscriptionId}/history	Dunning configuration and operational queues.
Wallet	BIL-05	GET /api/wallets/{subscriptionId}, GET /api/wallets/{subscriptionId}/balance, GET /api/wallets/{subscriptionId}/transactions?limit=50, GET /api/wallets/{subscriptionId}/topup-channels	Wallet view, balance, top-up channel visibility.

18.6 Fulfillment Operations APIs

Screen/action	Owning module/DD	Representative APIs/events	Frontend use
Order create/monitor	FUL-02 Order Capture	POST /api/orders, POST /api/orders/{id}/cancel, order query/detail APIs	Order capture and order monitor.
Order user tasks	FUL-02 KYC/payment/install steps	POST /api/orders/{id}/user-tasks/{task_id}/complete	Complete KYC/user tasks exposed by order flow.
Activation execution	FUL-03	POST /api/activation/execute, activation retry/admin APIs	Monitor/retry service activation execution.
Restriction execution	FUL-04	Restriction fulfillment APIs/events	Show network restriction/unrestriction execution state.
Termination execution	FUL-05	Termination fulfillment APIs/events	Show network termination execution state.
HomePass lifecycle	FUL-07, RLM-CFG-01	POST /api/homepasses/{id}/transitions/{process_id}/approve, HomePass detail APIs	Review HomePass lifecycle transitions.
Payment wait	BIL-01-PAY-01 event into FUL-02	PaymentApplied event consumed by FUL-02	Show waiting/received payment state.

Screen/action	Owning module/DD	Representative APIs/events	Frontend use
Install wait	WO-01 event into FUL-02	WorkOrderFinalized, WorkOrderCancelled	Show waiting/complete installation state.

18.7 Dispatch And Work Order APIs

Screen/action	Owning module/DD	Representative APIs	Frontend use
Work order list	WO-01	GET /api/work-orders?operatorCode=WIK&status=PENDING, ASSIGNED&kind=SUPPORT&techRegionCode=...&limit=50	Dispatch queue/list.
Work order detail	WO-01	GET /api/work-orders/{work_order_id}, GET /api/work-orders/{work_order_id}/print-pdf	Detail and print/download view.
Create work order	WO-01	POST /api/work-orders	Manual/support/admin WO creation where allowed.
Assignment	WO-01 + EM	POST /api/work-orders/{id}/assign, POST /api/work-orders/{id}/claim, EM contractor/staff lookup APIs	Assign/reassign/claim.
Notes/attachments	WO-01 + FILE_STORAGE	POST /api/work-orders/{id}/notes, POST /api/work-orders/{id}/attachments	Field/support evidence and notes.
Finalization	WO-01 flow satellites	POST /api/work-orders/{id}/finalize-first-confirm and flow-specific finalize APIs	Finalization review/confirmation.
Equipment binding	WO-01 + OSR	POST /api/work-orders/{id}/equipment-bindings	Capture serial/equipment binding from work order.
Cancellation	WO-01	POST /api/work-orders/{id}/cancel	Controlled cancellation.
Contractor/team lookup	EM-02	GET /api/contractors/{contractor_id}, GET /api/contractor-staff?contractorId=X&staffRoleTier=MAIN_HANDLER&active=true	Assignment candidate display.
Tech regions	ILM-CFG-02	GET /api/tech-regions?operatorCode=WIK&active=true, GET /api/tech-regions/{id}/candidate-group?taskType=INSTALL	Region/skill/candidate selection.

18.8 OSR And Equipment APIs

Screen/action	Owning module/DD	Representative APIs	Frontend use
Stock balances	OSR-01	GET /api/stock-balances?operatorCode=WIK&locationId=...	Stock view by location/SKU.
Stock movements	OSR-01	GET /api/stock-movements?operatorCode=WIK&locationId=X&skuId=Y&since=DATE, POST /api/stock-movements	Movement list and controlled movement creation.
Stock locations	OSR-01	GET /api/stock-locations/{location_id}	Location detail.
Stock reasons	OSR-01	GET /api/stock-reason-codes?operatorCode=WIK&active=true	Movement reason dropdowns.
Equipment instance detail	OSR-INSTANCE-01	GET /api/equipment-instances/{instance_id}	Serial/equipment detail.
Equipment instance search	OSR-INSTANCE-01	GET /api/equipment-instances?operator=WIK&locationId=...&state=IN_CONTRACTOR_STOCK	Search/list by state/location.
Equipment lifecycle	OSR-INSTANCE-01	GET /api/equipment-instances/{instance_id}/lifecycle-events, POST /api/equipment-instances/{instance_id}/lifecycle-events	Show/add lifecycle events where role allows.

Screen/action	Owning module/DD	Representative APIs	Frontend use
Swap/RMA	OSR-RMA-01	GET /api/swap-requests/{swap_request_id}, GET /api/swap-requests?operator=WIK¤tState=FAILED&since=..., POST /api/swap-requests/gpon, POST /api/swap-requests/hfc, POST /api/swap-requests/eqp, POST /api/swap-requests/equ	Swap/RMA queues and initiation.
Equipment catalog	PLM-CFG-06	GET /api/equipment-kinds?operatorCode=WIK&active=true, GET /api/equipment-skus?operatorCode=WIK&active=true&kindCode=ONT, GET /api/equipment-skus/{sku_id}	SKU/kind dropdowns and compatibility panels.

18.9 Catalog And Configuration APIs

Screen/action	Owning module/DD	Representative APIs	Frontend use
Service catalog	PLM-CFG-01	GET /api/services/by-code/{service_code}, service list/create/update APIs	Service catalog list/detail/edit.
Package management	SIP-01	GET /api/packages/{package_id}, package list/create/update APIs	Package catalog and offer configuration.
HomePass	RLM-CFG-01	GET /api/homepasses/{homepass_id}, POST /api/homepasses/bulk-import, HomePass transition APIs	HomePass search, import, lifecycle.
Tech regions	ILM-CFG-02	GET /api/tech-regions?operatorCode=WIK&active=true, GET /api/tech-regions/{id}, PATCH /api/tech-regions/{id}	Tech region admin.
Tax config	PLM-CFG-02	Tax rule/config APIs, POST /api/tax/admin/rules, POST /api/tax/compute	Tax admin and validation.
Wallet config	PLM-CFG-03	Wallet config APIs, wallet channel config APIs	Wallet behavior setup.
Discounts	PLM-CFG-04	Discount catalog APIs	Discount setup and activation.
Adjustment catalogs	PLM-CFG-05, PLM-CFG-08	GET /api/adjustment-reason-codes?operatorCode=WIK&active=true, POST /api/adjustment-reason-codes, PATCH /api/adjustment-reason-codes/{operator}/{reason_code}	Adjustment reason admin.
Equipment type catalog	PLM-CFG-06	GET /api/equipment-kinds, GET /api/equipment-skus, compatibility/fee APIs	Equipment kind/SKU/compatibility admin.
Billable event catalog	BIL-CFG-01	POST /api/billable-events, billable event catalog APIs	Chargeable event setup.

18.10 Workforce Admin APIs

Screen/action	Owning module/DD	Representative APIs	Frontend use
Contractor detail	EM-02	GET /api/contractors/{contractor_id}	Contractor profile.
Contractor staff	EM-02	GET /api/contractor-staff/{staff_id}, GET /api/contractor-staff?contractorId=X&staffRoleTier=MAIN_HANDLER&active=true	Staff list/detail and assignment candidates.
Contractor availability	EM-02	POST /api/em/contractor-availability	Availability update where role allows.

Screen/action	Owning module/DD	Representative APIs	Frontend use
Slot commitments	EM-02	GET /api/contractor-slot-commitments?slotId=X&date=Y, DELETE /api/contractor-slot-commitments/{id}	Slot/commitment visibility and admin cleanup.
Team/agent registry	EM-02 team/agent DD	Team/agent APIs from DD	Contractor team and agent mapping.

18.11 Platform Admin APIs

Screen/action	Owning module/DD	Representative APIs	Frontend use
Audit search	Audit/local module audit	GET /api/audit?... if central audit API is enabled; module audit APIs otherwise	Audit panel and compliance search.
Workflow incidents	Camunda foundation + module wrappers	Camunda incident/task endpoints through approved backend/admin proxy	Incident review and controlled retry.
Outbox replay	FOUNDATION_KAFKA/admin	POST /admin/outbox/replay	Replay events with platform role.
Inbox retry	FOUNDATION_KAFKA/admin	POST /admin/inbox/retry/{eventId}	Retry stuck consumer event.
Cache invalidate	FOUNDATION_CACHE/admin	POST /admin/cache/invalidate, GET /admin/cache/stats?keyPrefix=..., GET /admin/cache/reconcile?keyPrefix=...&sample=N	Cache admin and reconciliation.
Notification admin	NOT-01	POST /api/admin/notifications/{id}/resend, template/admin APIs	Notification recovery.
Invoice/payment admin	BIL admin APIs	POST /api/admin/invoices/{id}/rerender-pdf, POST /api/admin/payments/{id}/allocate-surplus	Controlled operational repair.

19. State And Data Loading

Rules:

- Use server-side pagination for tables.
- Use query parameters that match backend API contracts.
- Use optimistic UI only for low-risk local state; not for financial/subscription/equipment state.
- Refresh detail after a command completes.
- Use stale/error indicators per panel in aggregate screens.
- Avoid loading all modules at once if the screen can lazy-load panels.

Customer 360 loading strategy:

1. Load ILM customer/account first.
2. Render shell and identity panel.
3. Load subscriptions, billing, work orders, tickets, notifications in parallel.
4. Show panel-level loading/error states.
5. Allow retry per failed panel.

20. Command Handling

All create/update/action commands must:

- validate form locally;
- include correlation id;
- include idempotency key where retry/duplicate submit is possible;

- disable duplicate submit while in flight;
- show backend validation errors without losing form state;
- show final operation id for long-running commands;
- write no state directly to another module's local cache.

High-risk commands require:

- confirmation screen;
- reason/comment;
- impact summary;
- role check;
- audit visibility after completion.

21. Regional And Operator Configuration

Backoffice must consume FE-APP-00 runtime config.

Operator-specific examples:

Config	Example
Logo/theme	WIK Kenya logo and colors first; future WUG/WTZ/YASSN from config.
Language	en - KE first, future sw - KE, fr - SN, etc.
Currency	KES first, future UGX/TZS/XOF.
Tax labels	KRA labels for Kenya, future authority labels by country.
Feature flags	tax invoice, wallet, self-service actions, region-specific dunning behavior.
Address format	Kenya address/HomePass fields first, future regional formats by config.

Rules:

- Do not hardcode Kenya-only labels into shared components.
- Use translation keys for menu names, buttons, table headers, status labels, and errors.
- Show country/operator-specific legal text from runtime config where needed.

22. Security And Privacy

Backoffice exposes sensitive operational data.

Required controls:

- Auth through Keycloak/OIDC.
- Role-based navigation and action visibility.
- Backend authorization for every API.
- Session timeout and re-auth for sensitive actions where required.
- Mask sensitive PII unless role allows full view.
- Never log tokens, ID numbers, payment secrets, or raw document contents.
- Download files only through authorized APIs.
- Audit all high-risk actions.

Sensitive data examples:

- customer ID document number;
- KYC documents/photos;
- payment references;
- invoice/tax invoice documents;

- equipment serial/customer binding;
- internal notes;
- audit logs.

23. Accessibility And UX Rules

Backoffice must be efficient for repeated daily use.

Required:

- keyboard navigation for tables and forms;
- accessible labels and focus states;
- no color-only status;
- responsive behavior down to tablet width;
- clear empty/error/loading states;
- compact layout for operations;
- copy buttons for IDs/correlation ids;
- timestamps with timezone clarity.

Status colors must be consistent:

Status type	UI treatment
Success/completed	Green
Warning/pending/manual action	Amber
Error/failed/overdue	Red
Neutral/draft/inactive	Gray
Info/in-progress	Blue

24. Observability

Backoffice telemetry must include:

- app version;
- operator/country;
- route/screen id;
- user role family;
- command type;
- backend error code;
- correlation id;
- operation id when available;
- API latency buckets;
- panel load failures.

Do not send PII to analytics.

25. Implementation Notes

Recommended route structure:

```

/home
/customers
/customers/:customerId
/tickets
/tickets/:ticketId
/subscriptions

```

```
/subscriptions/:subscriptionId
/subscription-operations/:operationId
/billing/invoices
/billing/payments
/billing/dunning
/fulfillment/orders
/work-orders
/work-orders/:workOrderId
/equipment
/catalogs
/workforce
/admin/audit
/admin/incidents
/admin/outbox
/reports
```

Recommended shared components:

- DataTable
- SavedFilterBar
- StatusChip
- Timeline
- EntityHeader
- ActionMenu
- ConfirmActionDialog
- ImpactSummary
- AuditPanel
- AttachmentViewer
- TaskInbox
- PanelErrorBoundary
- CorrelationIdCopy

26. Acceptance Checklist

- [] Navigation is role-aware and feature-flag-aware.
- [] Customer 360 loads independent panels and handles partial backend failure.
- [] Subscription actions use preview/impact/confirm/operation-monitor flow.
- [] Billing high-risk actions require reason and confirmation.
- [] Work order assignment validates EM references.
- [] Equipment state changes call OSR APIs only.
- [] Catalog changes show lifecycle/effective-date impact.
- [] Global search routes to owning modules.
- [] All commands send correlation id.
- [] Retryable commands use idempotency key.
- [] Files download/upload through authorized APIs.
- [] Audit/correlation ids are visible for support.
- [] No Kenya-only text is hardcoded outside translation/config.